Journal of Affective Disorders 318 (2022) 167–174

Contents lists available at ScienceDirect

Journal of Affective Disorders

journal homepage: www.elsevier.com/locate/jad

Research paper

Cardiovascular biomarkers of response to accelerated low frequency repetitive transcranial magnetic stimulation in major depression

Jack Z. Sheen [a,b], Jean-Philippe Miron [a,c,g,h], Farrokh Mansouri [a], Katharine Dunlop [c,i,j], Thomas Russell [b], Ryan Zhou [b], Molly Hyde [a,b], Linsay Fox [b], Helena Voetterl [a], Zafiris J. Daskalakis [a,c,d,f], John D. Griffiths [a,c,e], Daniel M. Blumberger [a,c,d], Jonathan Downar [a,b,c,*]

[a] Institute of Medical Science, University of Toronto, Toronto, Canada
[b] Krembil Research Institute, University Health Network, Toronto, Canada
[c] Department of Psychiatry, Faculty of Medicine, University of Toronto, Canada
[d] Temerty Centre for Therapeutic Brain Intervention, Centre for Addiction and Mental Health, Canada
[e] Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health, Toronto, Canada
[f] Department of Psychiatry, University of California San Diego, San Diego, CA, USA
[g] Centre Hospitalier de l'Université de Montréal (CHUM), Centre de Recherche du CHUM (CRCHUM), Canada
[h] Département de Psychiatrie, Faculté de Médecine, Université de Montréal, Montréal, QC, Canada
[i] Keenan Research Centre for Biomedical Science, St. Michael's Hospital, Unity Health Toronto, USA
[j] Centre for Depression and Suicide Studies, St. Michael's Hospital, Unity Health Toronto, USA

ARTICLE INFO

Keywords:
Major depressive disorder
Low frequency rTMS
Electrocardiogram
Biomarkers

ABSTRACT

Background and objective: Repetitive transcranial magnetic stimulation (rTMS) is an effective and safe treatment for major depressive disorder (MDD). rTMS is in need of a reliable biomarker of treatment response. High frequency (HF) dorsolateral prefrontal cortex (DLPFC) rTMS has been reported to induce significant changes in the cardiac activity of MDD patients. Low frequency DLPFC rTMS has many advantages over HF-DLPFC rTMS and thus this study aims to further investigate the effect of low frequency 1 Hz right hemisphere (*R*)-DLPFC rTMS on the cardiac activity of MDD patients, as well as the potential of using electrocardiogram (ECG) parameters as biomarkers of treatment outcome.

Methods: Baseline ECG sessions were performed for 19 MDD patients. All patients then underwent 40 sessions of accelerated 1 Hz R-DLPFC rTMS one week after the baseline session.

Results: Heart rate (HR) significantly decreased from the resting period to the first and third minute of the 1 Hz R-DLPFC rTMS period. Resting HR was found to have a significant negative association with treatment outcome. Prior to Bonferroni correction, HR during stimulation and the degree of rTMS-induced HR reduction were significantly negatively associated with treatment outcome. No significant changes were observed for the heart rate variability (HRV) parameters.

Limitations: Sample size ($n = 19$); the use of electroencephalography equipment for ECG; lack of respiration monitoring; relatively short recording duration for HRV parameters.

Conclusion: This novel study provides further preliminary evidence that ECG may be utilized as a biomarker of rTMS treatment response in MDD.

Trial registration: ClinicalTrials.gov Identifier: NCT04376697.

1. Introduction

Major depressive disorder (MDD) is now the foremost cause of disability worldwide (Friedrich, 2017). Repetitive transcranial magnetic stimulation (rTMS) has entered increasingly widespread use as an evidence-based treatment for medication-resistant MDD. rTMS involves non-invasive brain stimulation using externally applied focused magnetic field pulses, which can induce lasting changes in the activity of

* Corresponding author at: MRI-Guided rTMS Clinic, Toronto Western Hospital, 7M-415 399 Bathurst Street, Toronto, ON M5T 2S8, USA.
E-mail address: jonathan.downar@utoronto.ca (J. Downar).

https://doi.org/10.1016/j.jad.2022.08.105
Received 23 December 2021; Received in revised form 4 July 2022; Accepted 26 August 2022
Available online 31 August 2022
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target brain regions (Rossi et al., 2009; Downar et al., 2014; Lipsman et al., 2014). Safety and tolerability have been well-studied and demonstrate minimal treatment-limiting adverse effects and very rare serious adverse effects (Rossi et al., 2009), while the efficacy of rTMS in MDD has been affirmed in several large meta-analyses (Brunoni et al., 2017; Lefaucheur et al., 2020; Milev et al., 2016). One recent meta-analysis has shown that rTMS of the dorsolateral prefrontal cortex (DLPFC) produces response and remission rates of up to 55 % and 35 % respectively in MDD (Milev et al., 2016).

Current objectives of translational rTMS research include accelerating treatment courses via multiple daily sessions and identifying reliable predictors of clinical response to rTMS prior to the start of a treatment course. Recent studies suggest that accelerated rTMS (arTMS) involving multiple treatment sessions per day are safe, tolerable, and may achieve similar outcomes in shorter timeframes of 4 to 10 days compared to the conventional timeframe of 30 days or more (McGirr et al., 2015; Baeken et al., 2013; Hadley et al., 2011; Holtzheimer et al., 2010; Cole et al., 2020). Predictive biomarkers, if proven reliable, could help to identify likely rTMS responders and non-responders, and could also provide objective measures for monitoring engagement of target circuits during treatment. Although clinical features have shown some group-level predictive power, it has been difficult to demonstrate reliable individual-level predictive power for any biomarker to date (Simon and Perlis, 2010; Bagby et al., 2002). The ideal biomarker would, in addition to having high individual-level predictive accuracy, be cost-effective and simple to acquire in real-world clinical settings.

Electrocardiography (ECG) is a convenient and low-cost technique used to measure cardiac activity via cutaneous changes in electrical potential, and due to the association between psychiatric disorders and cardiovascular abnormalities, ECG has emerged as a promising biomarker candidate of rTMS treatment outcome. MDD patients in particular has been shown to exhibit both an increase in resting heart rate (HR) and a decrease in heart rate variability (HRV) (Michael and Kaur, 2021), which are posited to occur as a result of reduced parasympathetic activity. Thus, MDD treatments that can increase parasympathetic activity may help to alleviate symptoms associated with abnormal cardiac activity. One reliable method of increasing parasympathetic drive and lowering HR is through electrical stimulation of the vagus nerve (VN), a major component of the parasympathetic branch of the autonomic nervous system (Buschman et al., 2006). VN stimulation has been shown to improve MDD symptoms, suggesting anatomical links to key neural substrates of mood (Rush et al., 2000). Anatomically, branches of the VN converge onto four medullary nuclei: the dorsal nucleus of VN (DVN), the nucleus ambiguus (AMB), the spinal trigeminal nucleus (SpV), and the nucleus of the solitary tract (NTS) (c.f. supplementary material of the referenced article) (Iseger et al., 2020a). In the control of HR, the NTS, DVN, and AMB all receive information from regions of the prefrontal cortex (Park and Thayer, 2014), including the DLPFC, the conventional target of rTMS treatment for MDD. Therefore, DLPFC rTMS could potentially engage target circuits for parasympathetic control of HR and successful target engagement could potentially be tracked via a simplified ECG montage in clinical settings.

Recent studies have shown that rTMS can indeed influence both HR and HRV (Michael and Kaur, 2021). A 2017 meta-analysis concluded that rTMS reduces HR and increases HRV to a moderate degree, and that the effects on HR and HRV were most pronounced for rTMS targeting the prefrontal cortex, including the DLPFC (Makovac et al., 2017). In healthy controls, Kaur et al. reported that low frequency rTMS influences HR differently from HF-rTMS (Kaur et al., 2020). Casanova et al. reported that low frequency rTMS reduced HR and increased multiple HRV parameters in autism spectrum disorder (ASD) patients post-treatment (Casanova et al., 2014). However, to date, only three studies have specifically examined the effect of rTMS on HR and HRV in MDD patients. Udupa et al. (2007) reported that 15 Hz left DLPFC rTMS did increase several HRV parameters; however, the increase was not correlated with MDD symptom improvement (Udupa et al., 2007). A follow-up study from the same group subsequently found that HF-rTMS reduced HR and increased multiple HRV parameters, and that the post-treatment ratio of low- to high-frequency power in the ECG recording (LF/HF ratio) correlated with treatment outcome (Udupa et al., 2011). More recently, Iseger et al. (2020b) found that, when compared to sham stimulation, active intermittent theta burst rTMS over the left DLPFC significantly decreased overall HR, increased HR deceleration over the first 30 to 60 s of the first treatment session and increased the following HRV parameters: low frequency (LF) power, high frequency (HF) power, the standard deviation of the RR intervals (SDNN), and the root mean square of the successive differences (RMSSD). The degree of increase in HR deceleration during the first 30 s of the first treatment session also showed a trend towards correlation with treatment outcome. This trend, however, did not reach statistical significance, possibly due the small size of the study population (Iseger et al., 2020b). Thus, currently available evidence suggests a possible group-level, if not an individual-level, relationship between changes in HR parameters and clinical response to left DLPFC rTMS.

The question remains whether these previous findings of HR changes with left HF-rTMS also apply to other treatment protocols such as low-frequency (1 Hz) right (*R*)-DLPFC rTMS, or to accelerated schedules involving multiple daily treatment sessions. Also of interest is whether any induced changes in HR and HRV with 1 Hz R-DLPFC rTMS may associate with treatment outcome. Compared to high frequency rTMS, low rTMS appears to have similar efficacy in MDD (Milev et al., 2016), a similar safety profile (Sun et al., 2012; Dobek et al., 2015), significantly better tolerability (Kaur et al., 2019), and potentially lower equipment cost, as well as proposed potential for eventual use in home settings (Miron et al., 2020). When considering the brain hemispheres, R DLPFC rTMS has shown very similar efficacy to the more commonly administered left or bilateral DLPFC rTMS (Milev et al., 2016; Fitzgerald et al., 2020). Importantly, compared to the left DLPFC, the right DLPFC has been proposed to exert a more inhibitory effect on cardiac output (c.f. supplementary material of the referenced article) (Iseger et al., 2020a). Therefore, the principal objective of the present study was to explore the effect of 1 Hz R-DLPFC rTMS on cardiac rhythm, and the association of this effect with arTMS treatment outcome. By advancing the current knowledge on the mechanism of action of rTMS in the treatment of MDD, we aim to further clarify the potential of ECG as an accurate, practical, and cost-effective biomarker.

Since the mechanistic differences between 1 Hz and HF-rTMS are still under investigation, we hypothesized that 1 Hz rTMS would significantly reduce HR during stimulation, based on results from previous studies examining the effect of high-frequency left DLPFC-rTMS on the cardiac rhythm of MDD patients (Iseger et al., 2020b). Furthermore, we hypothesized that the degree of reduction would have a significant positive association with treatment outcome for a subsequent course of accelerated (8 hourly sessions per day for 5 days) R-DLPFC rTMS in these patients. According to the results from previous studies on rTMS and cardiac rhythm in ASD patients (Casanova et al., 2014) and MDD patients (Udupa et al., 2007; Udupa et al., 2011; Iseger et al., 2020b), we also expected that HRV parameters would significantly change during 1 Hz R-DLPFC rTMS. Finally, based on other studies that have reported abnormal resting HR in MDD (Michael and Kaur, 2021), resting HR and HR during stimulation were hypothesized to have a significant negative association with treatment outcome.

2. Methods

2.1. Study design and participants

21 MDD participants, 12 female and 9 males, between the age of 28–61 were included in this single-arm, prospective, open-label feasibility study. Participants were included if they: 1) met criteria for MDD diagnosis (single or recurrent episode) on the Mini International Neuropsychiatric Interview (MINI); 2) maintained a stable medication

regimen from 4 weeks prior to the start of the study and could maintain a stable medication throughout the entire study period. Exclusion criteria were: 1) within the past 3 months, having a history of substance dependence or abuse; 2) having concomitant major and unstable medical illness; 3) having a cardiac pacemaker or implanted medication pump; 4) active suicidal intent; 5) as assessed by a study investigator, having a diagnosis of any primary personality disorder that is causing greater impairment than MDD; 6) diagnosis of any psychotic disorder; 7) any significant neurological disorder or insult (including, but not limited to: any condition likely to be associated with increased intracranial pressure, space-occupying brain lesion, any history of seizure confirmed diagnostically by neurological assessment (except those therapeutically induced by ECT), cerebral aneurysm, Parkinson's disease, Huntington's chorea, dementia, stroke, neurologically confirmed diagnosis of traumatic brain injury, or multiple sclerosis); 8) if participating in psychotherapy, must have been in stable treatment for at least 3 months prior to entry into the study (with no anticipation of change in the frequency of therapeutic sessions, or the therapeutic focus over the duration of the study); 9) in the opinion of the investigator, any clinically significant laboratory abnormality; 10) due to the potential to limit rTMS efficacy, a dose of more than lorazepam 2 mg daily (or equivalent) currently (or in the last 4 weeks) or any dose of an anticonvulsant; 11) any non-correctable clinically significant sensory impairment, and 12) any significant cardiovascular or metabolic disorder or insult including, but not limited to: coronary artery disease, abnormal heart rhythms, heart failure, cardiac valve disease, congenital heart disease, cardiomyopathy, vascular disease, dyslipidemia, diabetes, or hypertension. All participants provided informed consent and this study was approved by the Research Ethics Board of the University Health Network.

2.2. Cardiac and clinical outcome data acquisition

ECG acquisition was performed at the baseline session. As part of a 16-lead, 512 Hz EEG system (EasyCap, Brain Products, Munich, Germany), one electrode was attached to the left wrist of the participants to acquire ECG data. The FCz and AFz electrodes on the EEG cap were used as reference and ground electrodes, respectively. Stable impedances of <5 kΩ were acquired prior to the start of the recording session. Participants first underwent a 5-minute pre-stimulation resting state session, then a 10-minute 1 Hz R-DLPFC rTMS EEG/ECG session (1 single train, 600 pulses per session, 120 % of rMT), followed by another 5-minute post stimulation resting state session. All recordings were conducted from 9 am to 12 pm during the day.

arTMS treatment sessions took place one week after the baseline session. Clinical outcome was measured using the Beck Depression Inventory-II (BDI-II). Outcome scores used in this analysis were collected during the second follow-up session, which took place four weeks after the arTMS treatment session. This study focused specifically on the ECG data; results from the EEG recording component are to be the subject of a separate report.

2.3. arTMS treatment procedures

rTMS was delivered with a MagPro R20 stimulator equipped with a MC-B70 coil (MagVenture, Farum, Denmark). Resting motor threshold (rMT) was determined according to standard techniques (McClintock et al., 2017). Treatment consisted of an arTMS course of 8 hourly sessions per day over the 5 consecutive weekdays from Monday to Friday, totaling 40 sessions. Each rTMS session consisted of low-frequency 1 Hz stimulation delivered over a 10 min period (1 single train, 600 pulses per session, 50-minute inter-session interval) at 120 % of rMT over the right dorsolateral prefrontal cortex (R-DLPFC). DLPFC was located according to a previously published heuristic approximating the F4 EEG site using scalp measurements (Mir-Moghtadaei et al., 2017).

2.4. Cardiac data processing

Raw ECG data were visualized using the Python-based Plotly package (Plotly Technologies Inc, 2015). R-peaks of the QRS complex were visually detected and the time at which the peaks occurred were recorded to calculate the RR-interval (time lapse between two consecutive R-peaks), from which the HR was calculated in beats per minute. HRV was determined using the PyHRV package (Gomes, 2018). To minimize motion artifacts, 3 min from the end of the pre-stimulation resting period and the start of the post-stimulation resting period were used to calculate resting HR and HRV. The entire 10 min of recording during the 1 Hz R-DLPFC rTMS session were used to calculate HR and HRV during 1 Hz stimulation. Frequency-domain HRV parameters included in the analysis were very low frequency (VLF) power (0.0033–0.04 Hz), LF power (0.04–0.15 Hz), HF power (0.15–0.4 Hz), and the low frequency power to high frequency power ratio (LF/HF ratio). SDNN and RMSSD were the time domain parameters included in the analysis. To obtain normal distributions, a natural log transform was applied to all HRV parameters.

Of note, 2 out of 21 participants showed a pre-stimulation resting HR of 110.9 and 115.0 bpm respectively, exceeding the 100-bpm resting heart rate threshold of tachycardia (Custer and Rau, 2008). For the purposes of this analysis, we did not feel confident about the appropriateness of combining together results from patients with resting cardiac activity falling inside vs. outside medically accepted normal limits. Therefore, the decision to separate these 2 participants from the analysis of the 19 participants whose resting HR values fell within normal limits (60–100 bpm) was made a priori. Supplementary analyses with the inclusion of these 2 tachycardia participants are presented in Supplementary Figs. 2 and 3.

2.5. Statistical analysis

Due to the within-subjects design of the study, linear mixed effects models (LMMs) were conducted using the R (version 4.0.4) (R Core Team, 2021) package LME4 (Bates et al., 2015) to test for statistically significant differences between HR during the pre-stimulation resting period and each minute of the 10-minute 1 Hz R-DLPFC rTMS period, as well as the post-stimulation resting period. The 1 Hz R-DLPFC rTMS period was divided into ten 1-minute segments for analysis, since it was previously reported that theta burst rTMS exerted a significant effect on HR during the first minute of stimulation (Iseger et al., 2020b). Instead of averaging RR-intervals in each of the Sessions (pre-stimulation resting period, 1 Hz R-DLPFC rTMS period, and post-stimulation resting period), every RR-interval recorded in a Session were included in the models to more accurately account for the variance structure of the RR-intervals within each subject. The within-subject correlation of HR across the Sessions was accounted for by introducing random intercepts for each subject. The main fixed effects were defined as HR and Session, with every time-point of the RR-interval nested within Session. LMMs were also used to investigate whether HRV parameters (averaged over 3 min for the pre-stimulation resting period and over the entire 1 Hz R-DLPFC rTMS period) were significantly different across the sessions. The Bonferroni correction was used to adjust *p*-values for multiple comparisons (Bland and Altman, 1995). Where applicable, both the uncorrected p-value and corrected p-value are presented for ease of interpretation, since the Bonferroni correction increases the risk of Type II errors (Garamszegi, 2006). Linear regression models from R (R Core Team, 2021) were used to investigate the association between the mean HR during the pre-stimulation resting period and BDI-II percent improvement after the arTMS treatment, the association between mean HR during the 1 Hz R-DLPFC rTMS period and BDI-II percent improvement after the arTMS treatment, as well as the association between the reduction in HR from the pre-stimulation resting period to the 1 Hz R-DLPFC rTMS period and BDI-II percent improvement after the arTMS treatment.

3. Results

3.1. Demographics

As stated previously, data from 19 out of 21 participants were included in the analysis. The mean age in years of the study population was 41.2 ± 10.4 SD. Average baseline BDI-II was 35.5 ± 10.9. Full demographic and clinical characteristics are given in Table 1.

Among these 19 remaining patients, BDI scores improved from 35.5 ± SD 10.09 at baseline to 22.9 ± SD 11.3 on the final treatment day and 20.9 ± SD 14.5 at 4 weeks post-treatment. The response rate 4 weeks after treatment was 57.9 %. The remission rate 4 weeks after treatment was 42.1 %. Full clinical outcomes are summarized in Table 2.

3.2. Effect of 1 Hz R-DLPFC rTMS on HR

To investigate the effect of 1 Hz R-DLPFC rTMS on HR, HR during the pre-stimulation resting period was compared to HR during each minute of the rTMS period, as well as to HR during the post-stimulation resting period, using LMMs. Compared to the mean HR during the pre-stimulation resting period, the mean HR during the first 3 min of the 1 Hz R-DLPFC rTMS period was reduced by 3.32 bpm ($F(1, 2919) = 8.7260$, $p = 0.0032$ uncorrected, $p = 0.032$ corrected), 2.67 bpm ($F(1, 2989) = 6.6601$, $p = 0.0099$ uncorrected, $p = 0.09$ corrected), and 2.20 bpm ($F(1, 3035) = 11.4063$, $p = 0.0007$ uncorrected, $p = 0.007$ corrected) respectively. After an initial reduction, the mean HR of the 1 Hz R-DLPFC rTMS period and post-stimulation resting period steadily returned to near-baseline values over the 10 min period of 1 Hz stimulation (Fig. 1). The full statistical summary of the effects of 1 Hz rTMS on HR over this period is presented in Table 3.

3.3. Effect of 1 Hz R-DLPFC rTMS on HRV

Of the HRV parameters tested (VLF, LF, HF, LF/HF ratio, SDNN, and RMSSD), none were found to be significantly different between the pre-treatment resting period and the 1 Hz R-DLPFC rTMS period; LMMs were used for the comparison (Supplementary Fig. 1). The association between the resting HRV parameters and BDI-II percent improvement was assessed using linear regression. Only the resting LF/HF ratio was found to have a significant negative association with BDI-II percent improvement; $p = 0.0142$ uncorrected, $p = 0.1278$ corrected, $R^2 = 0.265$ (Supplementary Figs. 5 and 6).

3.4. HR and treatment outcome

Mean HR during the pre-stimulation resting period showed a significant negative association with BDI-II percent improvement after the arTMS treatment ($p = 0.0039$ uncorrected, $p = 0.0117$ corrected) (Fig. 2a.), with BDI-II percent improvement decreasing by 1.9 % for every bpm increase in the mean HR during the pre-stimulation resting period (slope = −1.9179), and 39.6 % of the variance in BDI-II percent improvement being explained by the model containing only the mean HR during the pre-stimulation resting period as the independent variable ($R^2 = 0.3960$).

Regarding HR during stimulation, we first assessed for associations between the absolute HR and subsequent treatment outcome (Fig. 2b). Since the third minute during the 1 Hz R-DLPFC rTMS period was found to have the most significant within-subject reduction in HR from the pre-stimulation resting period after Bonferroni correction, the mean HR during the third minute of the 1 Hz R-DLPFC rTMS period was selected as the independent variable for linear regression models involving the 1 Hz R-DLPFC rTMS period. 25.2 % of the variance in BDI-II percent improvement was explained by the model containing only the mean HR during the third minute of the 1 Hz R-DLPFC rTMS period as the independent variable ($R^2 = 0.2524$). Lower HR during the third minute of 1 Hz rTMS significantly associated with greater subsequent improvement in symptoms after treatment; however, this relationship no longer reached significance after Bonferroni correction ($p = 0.0283$ uncorrected, $p = 0.0849$ corrected).

We next tested for associations between the *reduction* in mean HR from baseline to the third minute of the 1 Hz rTMS period and subsequent treatment outcome (Fig. 2c). 16 out of 19 patients (84.2 %) showed a reduction in mean HR from the pre-stimulation resting period to the third minute of the 1 Hz rTMS period. This reduction in mean HR showed a significant negative association with BDI-II percent improvement after the arTMS treatment ($p = 0.0397$ uncorrected). 26.3 % of the variance in BDI-II percent improvement was explained by the model containing only the reduction in mean HR from the pre-stimulation resting period to the third minute of the 1 Hz R-DLPFC rTMS period as the independent variable ($R^2 = 0.2632$). Thus, contrary to our initial hypothesis, patients with *less* reduction in HR during stimulation showed a *greater* subsequent improvement in symptoms after treatment. However, once again, this relationship no longer reached significance after Bonferroni correction ($p = 0.0397$ uncorrected, $p = 0.1191$ corrected). No statistically significant association was found between resting HR and reduction in HR (from the pre-treatment resting period to the third minute of the 1 Hz rTMS stimulation period) (Supplementary Fig. 7).

Table 1
Demographic and clinical characteristics ($n = 19$).

Characteristics	No. (St.dev)
Age, years	41.2 (10.4)
Women	57.89 %
Age of onset, years	20.2 (10.0)
Length of current depressive episode, months	12.2 (12.3)
Baseline BDI-II	35.5 (10.9)
Antidepressant	57.9 %
Antidepressant combination	15.8 %
Antipsychotic augmentation	10.5 %
Psychostimulant augmentation	26.3 %
Benzodiazepine	5.3 %
ATHF total score	6.7 (5.4)
ATHF number of trials, current episode	1.3 (1.4)
ATHF highest score	3.2 (1.3)

Abbreviations: BDI-II = Beck Depression Inventory-II, ATHF = Antidepressant Treatment History Form.

Table 2
Clinical outcomes of 1 Hz R-DLPFC arTMS (n = 19).

Outcome measure: BDI-II	No. (St.dev)
Score baseline	35.5 (10.09)
Score on last treatment day	22.9 (11.3)
Score 1 week after treatment	21.7 (11.6)
Score 4 weeks after treatment	20.9 (14.5)
Percent improvement on last treatment day	34.2 % (29.0 %)
Percent improvement 1 week after treatment	39.4 % (25.8 %)
Percent improvement 4 weeks after treatment	40.0 % (35.7 %)
Response rate (>50 % improvement) after last treatment day	31.6 % (6/19)
Response rate (>50 % improvement) 1 week after treatment	42.1 % (8/19)
Response rate (>50 % improvement) 4 weeks after treatment	57.9 % (11/19)
Remission rate (≤12 on BDI-II) after last treatment day	10.5 % (2/19)
Remission rate (≤12 on BDI-II) 1 week after treatment	15.8 % (3/19)
Remission rate (≤12 on BDI-II) 4 weeks after treatment	42.1 % (8/19)

Abbreviations: BDI-II = Beck Depression Inventory-II.

4. Discussion

This study assessed, for the first time, the effects of 1 Hz R-LPFC rTMS on cardiac activity among MDD patients undergoing an accelerated regimen of treatment, as well as the relationship between such effects and the subsequent treatment outcome. As hypothesized, HR was significantly reduced from the pre-stimulation resting period to the first, second, and third minute of 1 Hz R-DLPFC rTMS, in a manner similar to

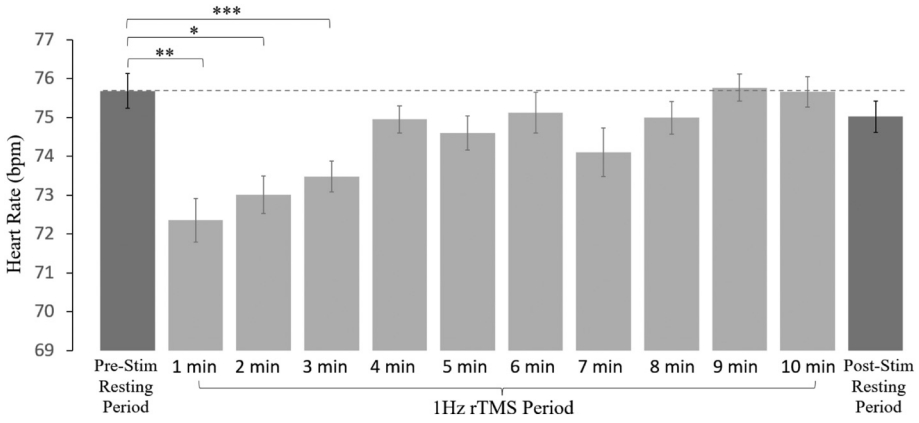


Fig. 1. Effect of 1 Hz R-DLPFC rTMS on HR. First and last bars represent the mean HR over the pre-stimulation resting session and the post-stimulation resting session. Light grey bars represent the mean HR during each minute of the 10-minute 1 Hz R-DLPFC rTMS session. Error bars represent standard errors of the mean. HR during the pre-stimulation resting period was compared to HR during each minute of the rTMS period, as well as to HR during the post-stimulation resting period. Compared to the mean HR during the pre-stimulation resting period, the mean HR during the first 3 min of the 1 Hz R-DLPFC rTMS period was reduced by 3.32 bpm, 2.67 bpm, and 2.20 bpm respectively. Uncorrected: * $p = 0.0099$, ** $p = 0.0032$, *** $p = 0.0007$; corrected: $p = 0.099$, $p = 0.032$, $p = 0.007$. The dashed line represents the mean HR during the pre-stimulation resting period, and demonstrates that, after an initial reduction, the mean HR of the 1 Hz R-DLPFC rTMS period and post-stimulation resting period eventually returned to nearly identical values as that of the pre-stimulation resting period.

Table 3
Summary of the comparison of HR during the pre-stimulation resting session to HR during each minute of the 1 Hz rTMS-EEG/ECG period and post-stimulation resting period. Statistical results from linear mixed effects models are reported.

Session	Mean HR, BPM (St. error)	Statistical comparisons (vs. pre-stimulation resting)	Bonferroni corrected p value
Pre-stimulation resting	75.68 (0.42)	N/A	N/A
1 Hz rTMS 1 min	72.36 (0.56)	F = 8.7260, df = 1, df.res = 2919, p = 0.0032	0.0320
1 Hz rTMS 2 min	73.01 (0.48)	F = 6.6601, df = 1, df.res = 2989, p = 0.0099	0.0990
1 Hz rTMS 3 min	73.48 (0.40)	F = 11.4063, df = 1, df.res = 3035, p = 0.0007	0.0070
1 Hz rTMS 4 min	74.95 (0.35)	F = 1.0249, df = 1, df.res = 3190, $p = 0.3235$	1.0000
1 Hz rTMS 5 min	74.60 (0.44)	F = 1.7420, df = 1, df.res = 3188, $p = 0.2018$	1.0000
1 Hz rTMS 6 min	75.12 (0.52)	F = 0.4247, df = 1, df.res = 3221, $p = 0.5220$	1.0000
1 Hz rTMS 7 min	74.10 (0.62)	F = 1.7825, df = 1, df.res = 3153, $p = 0.1968$	1.0000
1 Hz rTMS 8 min	74.99 (0.42)	F = 0.7156, df = 1, df.res = 3217, $p = 0.4076$	1.0000
1 Hz rTMS 9 min	75.77 (0.35)	F = 0.0191, df = 1, df.res = 3351, $p = 0.8915$	1.0000
1 Hz rTMS 10 min	75.66 (0.38)	F = 0.0064, df = 1, df.res = 3291, $p = 0.9370$	1.0000
Post-stimulation resting	75.01 (0.39)	F = 0.4578, df = 1, df.res = 3228, $p = 0.5064$	1.0000

the reductions previously observed for high-frequency or theta-bust rTMS of the left DLPFC (Iseger et al., 2020b). Regarding the relationship to treatment outcomes, HR during the pre-treatment resting period had a significant negative association with 1 Hz R-DLPFC arTMS treatment outcome. Patients with lower HR during the third minute of the 1 Hz R-DLPFC rTMS period showed greater improvement following 1 Hz R-DLPFC arTMS treatment outcome, but this relationship did not reach statistical significance after Bonferroni correction. Furthermore, unexpectedly, this association was weaker than the association between resting (pre-treatment) HR and treatment outcome ($p = 0.0039$ compared to $p = 0.0283$ before correction). Although it was hypothesized that the stimulatory effect of 1 Hz R-DLPFC rTMS on the parasympathetic nervous system would induce a stronger association between HR and outcome, the results did not support this prediction. In addition, and again contrary to the original hypothesis, patients with *less* of a decrease in HR during 1 Hz rTMS showed a *greater* subsequent improvement in symptoms; although again, this relationship no longer reached significance after Bonferroni correction. Finally, and once more contrary to our prediction, no statistically significant difference was found between the HRV parameters during the pre-stimulation resting state and the HRV parameters during the 1 Hz R-DLPFC rTMS period. Out of all resting HRV parameters, only the resting LF/HF ratio was found to have a significant negative association with BDI-II percent improvement prior to Bonferroni correction.

The HR reducing effect of rTMS has previously been reported for high frequency left DLPFC rTMS in MDD patients (Udupa et al., 2011; Iseger et al., 2020b) and for low frequency 0.5 Hz R-DLPFC rTMS in autism spectrum disorder patients (Casanova et al., 2014). Kaur et al. have also investigated the effect of low versus high frequency rTMS on HR in healthy controls (Kaur et al., 2020). Nonetheless, the present study is the first known report of low frequency rTMS reducing HR in MDD patients. The degree of HR reduction during 1 Hz R-DLPFC rTMS, (3.32 bpm, 2.67 bpm, and 2.20 bpm for the first, second, and third minute of stimulation respectively), was smaller than the previously reported 8.36 bpm reduction during the first minute of intermittent theta burst left DLPFC-rTMS, but larger than the 1.85 bpm reduction previously reported for 10 Hz left DLPFC-rTMS (Iseger et al., 2020b). Also, the reduction in HR was consistently observed across 84.2 % of the study participants. As previously hypothesized, this reduction in HR may be mediated through the interaction between the prefrontal cortex and the medullary nuclei onto which the branches of the vagus nerve converge (Park and Thayer, 2014). rTMS of the R-DLPFC may ultimately result in an increase in vagal tone and thereby increase parasympathetic activity. However, it is possible, and thus remains an area of control for future studies, that the reduction in HR may be the result of other physiological impact of rTMS, such as induced pain. Albeit rTMS-induced pain is more likely to increase sympathetic activity and increase HR (Hamunen et al., 2012) instead of decreasing HR. rTMS-induced pain may also increase respiration rate, which can potentially increase HR as well (Perry et al., 2019). Whatever mechanisms are in place, it is of interest to note that HR appears to decrease both for high-frequency/theta burst left DLPFC rTMS and for low-frequency R-DLPFC rTMS, in a manner paralleling the similar therapeutic outcomes in MDD for these two forms of stimulation (Fitzgerald et al., 2020). It should also be noted that the gradual return of HR during the 1 Hz R-DLPFC rTMS period to near baseline levels after an initial reduction (Fig. 1) is consistent with the fast speed of parasympathetic response (Iseger et al., 2020b) and is indicative that the effect of 1 Hz R-DLPFC rTMS on HR may be transient and most pronounced during the initial onset of stimulation. This suggests that, in contrast to lengthy EEG and MRI

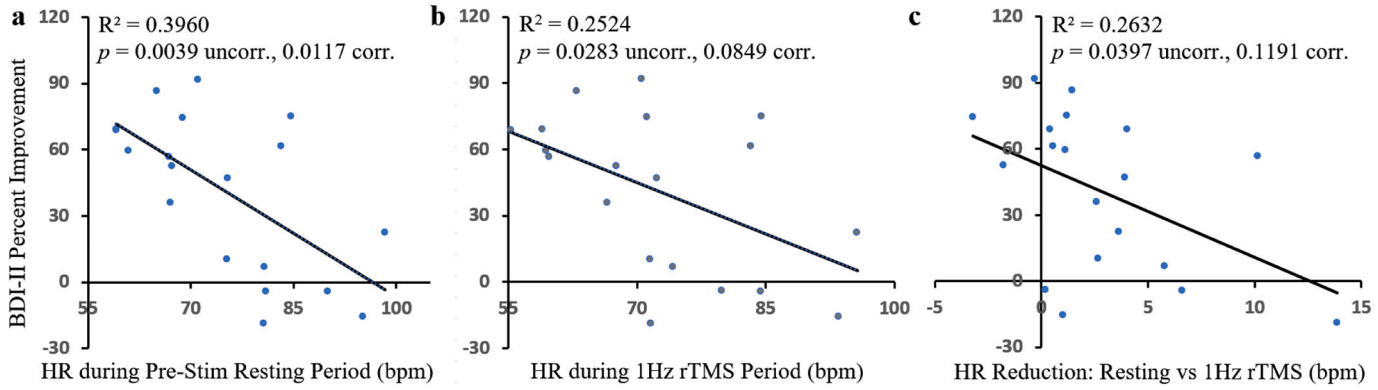


Fig. 2. Relationship between the mean heart rate and BDI-II percent improvement. Each dot represents one subject. (a) Linear regression showed a significant negative association between the mean heart rate during the pre-stimulation resting period and BDI-II percent improvement ($p = 0.0039$ uncorrected, $p = 0.0117$ corrected). 39.6 % of the variance in BDI-II percent improvement was explained by the model containing only the mean heart rate during the pre-stimulation resting period as the independent variable ($R^2 = 0.3960$). BDI-II percent improvement decreased by 1.9179 % for every bpm increase in the mean heart rate during the pre-stimulation resting period (slope = −1.9179). (b) Before Bonferroni correction, linear regression showed a significant negative association between the mean heart rate during the third minute of the 1 Hz R-DLPFC rTMS period and BDI-II percent improvement ($p = 0.0283$ uncorrected, $p = 0.0849$ corrected). 25.2 % of the variance in BDI-II percent improvement was explained by the model containing only the mean heart rate during the third minute of the 1 Hz R-DLPFC rTMS period as the independent variable ($R^2 = 0.2524$). BDI-II percent improvement decreased by 1.5522 % for every bpm increase in the mean heart rate during the third minute of the 1 Hz R-DLPFC rTMS period (slope = −1.5522). (c) Before Bonferroni correction, linear regression indicated a significant negative association between the reduction in mean heart rate from the pre-stimulation resting period to the third minute of the 1 Hz R-DLPFC rTMS period and BDI-II percent improvement ($p = 0.0397$ uncorrected, $p = 0.1191$ corrected). 26.32 % of the variance in BDI-II percent improvement can be explained by the model containing only the reduction in mean heart rate from the pre-stimulation resting period to the third minute of the 1 Hz R-DLPFC rTMS period as the independent variable ($R^2 = 0.2632$). BDI-II percent improvement decreased by 4.468 % for every bpm increase in the reduction in mean heart rate from the pre-stimulation resting period to the third minute of the 1 Hz R-DLPFC rTMS period (slope = −4.468).

sessions, only brief 3-minute sessions may be necessary when utilizing ECG as a biomarker of rTMS treatment outcome.

The negative association between HR during the pre-treatment resting period and BDI-II percent improvement is indicative that baseline cardiac activity may have an important role in the outcome of rTMS treatment in MDD. Having an elevated HR could be the result of autonomic nervous system disturbance that reduces that therapeutic benefit of rTMS. The unexpectedly weaker significance of the relationship between HR during actual stimulation (i.e., during the third minute of the rTMS period) and subsequent symptom improvement requires further investigation in studies with a larger population. To the best of our knowledge, this is the first suggestion of any association between pre-treatment resting HR and rTMS treatment outcome in MDD patients; however, as the *p*-value did not remain significant after Bonferroni correction, replication would be required in a larger sample.

Furthermore, the mildly significant association between *lesser* reductions in HR during 1 Hz R-DLPFC rTMS and *greater* clinical improvement is in apparent contradiction to previously reported results from Iseger et al. (2020b), where greater HR deceleration correlated with greater clinical improvement (Iseger et al., 2020b). However, some notable differences exist between the methods of the present study and Iseger 2020. First, Iseger et al. (2020b) examined the effect of high frequency intermittent theta burst (iTBS) rTMS on HR, while the present study utilized low frequency 1 Hz rTMS (Iseger et al., 2020b). It was originally hypothesized that the two protocols would impact cardiac rhythm in MDD similarly. However, the two stimulation patterns have been previously shown to impact cortical excitation and motor evoked potentials oppositely (Fitzgerald et al., 2006), which could potentially contribute to the opposing findings between the present study and those of Iseger et al. (2020b). Moreover, the site of stimulation in Iseger et al. (2020b) was the left DLPFC (Iseger et al., 2020b) rather than the R-DLPFC as in the present study. The left and right DLPFC have been suggested to differentially influence cardiac rhythm (c.f. supplementary material of the referenced article) (Iseger et al., 2020a). Lastly, the trending relationship between the increase in HR deceleration and treatment outcome described by Iseger et al. (2020b), irrespective of correction for multiple comparisons, did not reach statistical significance, while the R^2 value of 0.2570 used to indicate the potential trends (Iseger et al., 2020b) is almost identical to that of the current results ($R^2 = 0.2632$), indicating that the relationship described in Iseger et al. (2020b) is statistically milder than the association described in the current study. Thus, considering the dissimilarities between the two studies, it may not be unreasonable to theorize that the direction of association or trend between rTMS-induced HR reduction and treatment outcome may differ between the two studies. Since the etiology of MDD is complex, the relationship between HR reduction and the lesser treatment response observed in the current study may suggest that participants with a higher degree of HR reduction have comorbidities that are not well-addressed by 1 Hz arTMS, despite the potentially beneficial increase in parasympathetic activity.

Previous studies have reported that rTMS in MDD resulted in significant changes in multiple HRV parameters (Udupa et al., 2007; Udupa et al., 2011; Iseger et al., 2020b). In the present study, however, we found no statistically significant changes in HRV parameters between pre-stimulation resting period and the 1 Hz R-DLPFC rTMS period. One potential explanation for this discrepancy may be our use of relatively short recording sessions, especially for the 3-minute pre-stimulation resting period. Longer recording sessions may be required to more accurately measure all HRV parameters of interest (Shaffer and Ginsberg, 2017). Alternatively, these measures may indeed prove not to be robustly affected by 1 Hz rTMS and have no robust relationship with clinical outcome. In addition, the resting LF/HF ratio was found to have a significant negative association with BDI-II percent improvement. Since increase in the LF/HF ratio can be interpreted as a reduction in parasympathetic drive (Michael and Kaur, 2021), this result suggests that patients with higher parasympathetic drive during the resting period may improve more on 1 Hz R-DLPFC arTMS, congruent with the significant negative association found between resting HR and treatment outcome. However, the association was no longer significant after Bonferroni correction and the accuracy of this result may also be affected by our shorter recording sessions. Also, a previous report found no association between HRV parameters and treatment outcome for HF rTMS (Udupa et al., 2007).

This preliminary study has several limitations that future studies

should consider addressing. First and foremost, having a double-blinded placebo-controlled design could have substantially improved the generalizability of our results by isolating the effect of rTMS from other potential confounds. Second, our small sample size may have been inadequate to provide sufficient statistical power for some of the linear regression models to overcome Bonferroni correction. The sample size may also have been too small to effectively control for meaningful covariates that may affect HR such as sleep, caffeine use, exercise, and alcohol consumption. Third, ECG data were recorded using EEG equipment with a single electrode at the wrist, which is practical and convenient but may lack in accuracy compared to the standard ECG equipment and other monitoring methods such as chest monitoring. Moreover, the lack of control for respiration during ECG recordings, which can have a profound impact on HRV (Aysin and Aysin, 2006) and HR (Perry et al., 2019), may be problematic. It is possible the HR reduction seen during the first few minutes of rTMS stimulation could have been the result of participants anticipating the initiation of the first pulse and exercising relaxation through means such as respiration control. However, since our study recruited rTMS naïve patients and they were not instructed to practice active relaxation prior to the first stimulation session, most patients are more likely to be anxious prior to experiencing their first stimulation session, resulting in increased respiration rate and HR. The extent of such effects on HR can be further elicited by respiration and anxiety monitoring, both of which are lacking in the current report and have not yet been addressed, to our knowledge, in any studies published to date on the effect of rTMS on cardiac rhythm in MDD (Udupa et al., 2007; Udupa et al., 2011; Iseger et al., 2020b). Finally, the 3-minute recording length of the pre-stimulation resting period ECG sessions may also not have been adequate for accurate comparison of HRV parameters (Shaffer and Ginsberg, 2017).

To conclude: in agreement with previous reports on HF-rTMS, 1 Hz R-DLPFC rTMS pulses also appear to reduce HR. Our results also show a relationship of Baseline HR to treatment outcome in the case of 1 Hz right DLPFC-rTMS. However, the magnitude of HR reduction from the resting period to the 1 Hz R-DLPFC rTMS period, as well as HR during 1 Hz stimulation, were only moderately associated with treatment outcome, and did not remain significant after Bonferroni correction. No HRV parameters significantly changed from the resting period to the 1 Hz R-DLPFC rTMS period. Taking these findings together with the previous literature, the emerging picture is that HR may decrease similarly across multiple protocols used in MDD, across both left and right DLPFC. However, to date, the relationship between this effect and subsequent treatment outcome is less consistent in either magnitude or direction, and so far, does not appear to be particularly strong even at the group level, let alone at the desired level of predicting outcome for individual patients. Future work implementing a double-blinded placebo-controlled design, larger sample sizes, standardized ECG acquisition methods and equipment, respiration and anxiety monitoring, and potentially more refined techniques to ensure target engagement during treatment (i.e., image guidance and/or personalized targeting (Cole et al., 2020)), could help to further determine whether or not HR reduction can be used reliably to predict rTMS treatment outcome at the individual level.

Funding

This work was supported in part by the Toronto General and Western Hospital Foundation, the Arrell Family Foundation, the Brain & Behavior Research Foundation and the Branch Out Neurological Foundation.

CRediT authorship contribution statement

JZS was responsible for data collection, analysis, and wrote the manuscript. **JPM** led and designed the study and reviewed the manuscript. **FM** and **KD** assisted in the analysis and reviewed the manuscript. **TR**, **RZ**, **MH**, **LF**, **HV** participated in study design, data collection, and reviewed the manuscript. **ZJD** supported the study and reviewed the manuscript. **JDG**, **DMB** and **JD** supervised the study and reviewed the manuscript.

Conflict of interest

JZS, **FM**, **KD**, **TR**, **RZ**, **MH**, **LF**, **HV** do not report any conflict of interest.

DMB receives research support from CIHR, NIH, Brain Canada and the Temerty Family through the CAMH Foundation and the Campbell Research Institute. He received research support and in-kind equipment support for an investigator-initiated study from Brainsway Ltd. and he has been the site principal investigator for sponsor-initiated studies for Brainsway Ltd. He also receives in-kind equipment support from Magventure for investigator-initiated studies. He received medication supplies for an investigator-initiated trial from Indivior.

ZJD has received research and equipment in-kind support for an investigator-initiated study through Brainsway Inc. and Magventure Inc. His work has been supported by the Canadian Institutes of Health Research (CIHR), the National Institutes of Mental Health (NIMH), Brain Canada and the Temerty Family and Grant Family and through the Centre for Addiction and Mental Health (CAMH) Foundation and the Campbell Institute.

JD has received research support from the Arrell Family Foundation, the Buchan Family Foundation, Brain Canada, the Canadian Biomarker Integration Network in Depression, the Canadian Institutes of Health Research (CIHR), the Klarman Family Foundation, NIH, the Ontario Brain Institute, the Toronto General and Western Hospital Foundation, and the Weston Family Foundation; he has received travel stipends from Lundbeck and ANT Neuro; he has served as an advisor for BrainCheck, Restorative Brain Clinics, and TMS Neuro Solutions.

JPM reports research grants from the Brain & Behavior Research Foundation (BBRF) Young Investigator Award and the Réseau Québécois sur le Suicide, les troubles de l'Humeur et les troubles Associés (RQSHA) as well as salary support for his graduate studies from the Branch Out Neurological Foundation.

JDG has received research support from the Labatt Family Network, the CAMH Discovery Fund, the University of Toronto EMHSeed program, and the Tri-Council Canada-UK AI Initiative.

Acknowledgements

We would like to sincerely thank Jane Lee, Jianjun Shen, Terri Cairo, Julian Kwok, Meaghan Todd, Nuno Ferreira, Igbinosa Uwadiae, Michael Aiello, Vanathy Niranjan, Duong Nguyen, Sandy Ma, Peter Fettes, Laura Schulze, Arsalan Moghtadaei, Sunny Hong, Ekaterina Kosyachkova, and Eileen Lam for their most valued support, without which this study would not have been possible. We would also like to express our heartfelt appreciation for the trust, patience, and dedication of all the participants in this study. JPM would like to thank the Brain & Behavior Research Foundation and the Branch Out Neurological Foundation for their financial support of this project.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2022.08.105.

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